

CELL CHEMISTRY AND BIOENERGETICS

CHAPTER 2

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MBC-A course 11.09.2026



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The cell as a living industrial city

resources,
raw materials



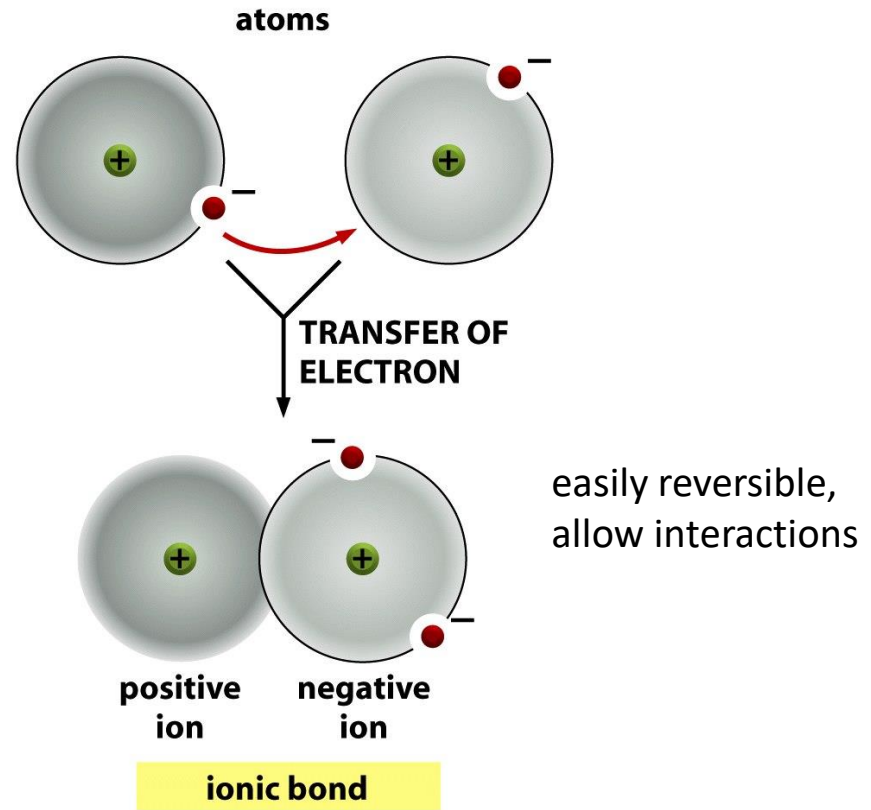
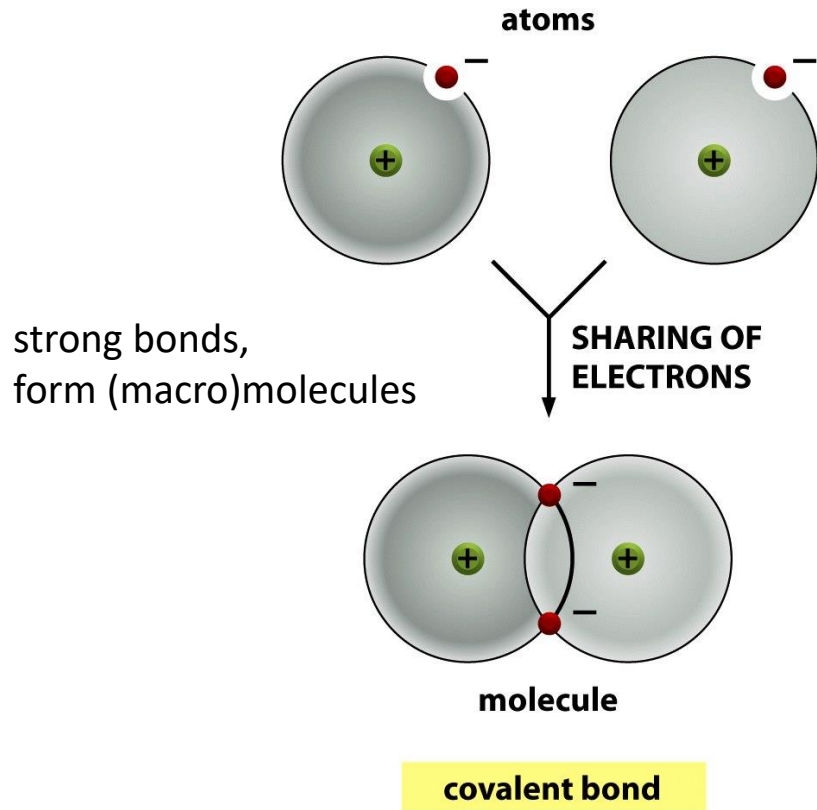
energy
production

construction and
maintenance

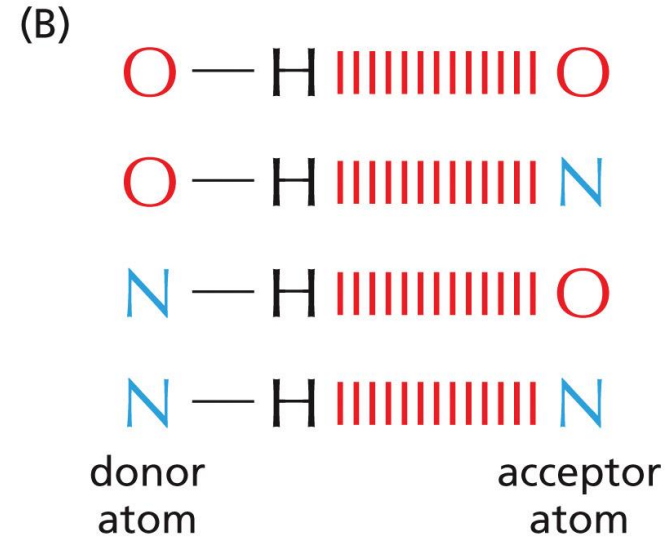
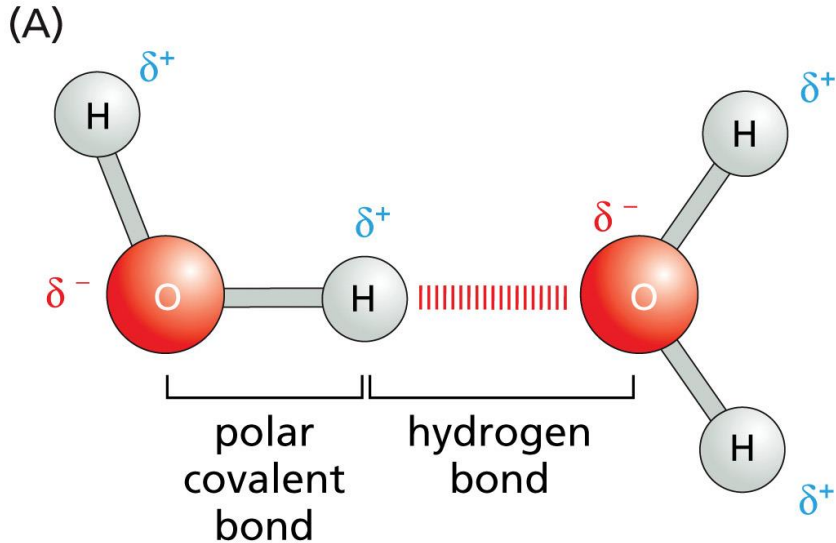


waste

It's all about bonds...



Hydrogen bonds form between H and electronegative atoms



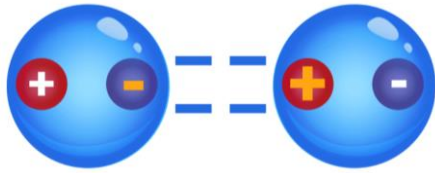
Weak interactions, very large numbers! They keep water a liquid at RT

→ polar/charged molecules easily dissolve in water (*hydrophilic*)

Weak interactions also occur in non-polar molecules

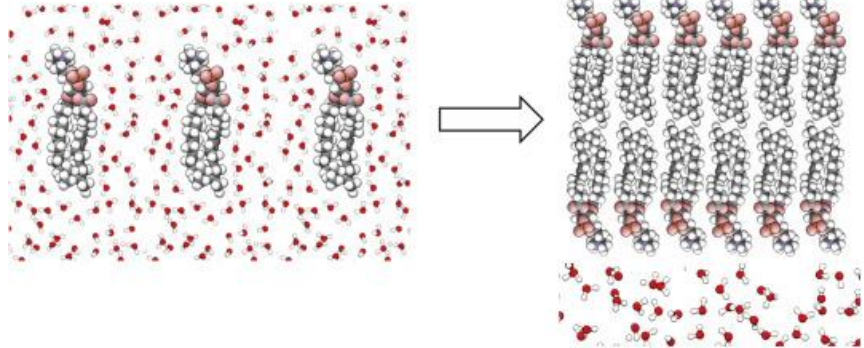
Van der Waals forces

weak attraction between
transient charges in atoms



Hydrophobic force

Aggregation of hydrophobic
molecules due to forced exclusion
from a polar environment



Noncovalent attractions allow molecules to interact

TABLE 2-1 Covalent and Noncovalent Chemical Bonds

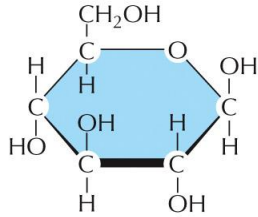
Bond type		Length (nm)	Strength kJ/mole**	
			in vacuum	in water
Covalent		0.15	377 (90)	377 (90)
Noncovalent	ionic*	0.25	335 (80)	12.6 (3)
	hydrogen	0.30	16.7 (4)	4.2 (1)
	van der Waals attraction (per atom)	0.35	0.4 (0.1)	0.4 (0.1)

*An ionic bond is an electrostatic attraction between two fully charged atoms. **Values in parentheses are kcal/mole. 1 kJ = 0.239 kcal and 1 kcal = 4.18 kJ.

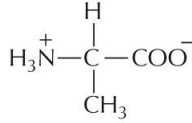
Table 2-1 Molecular Biology of the Cell 6e (© Garland Science 2015)

+ hydrophobic force

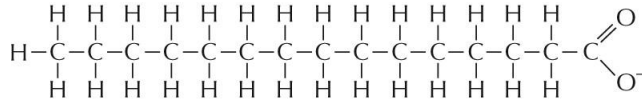
Main families of small organic molecules



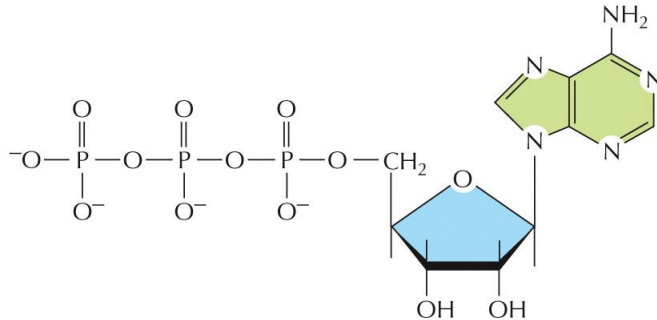
A SUGAR



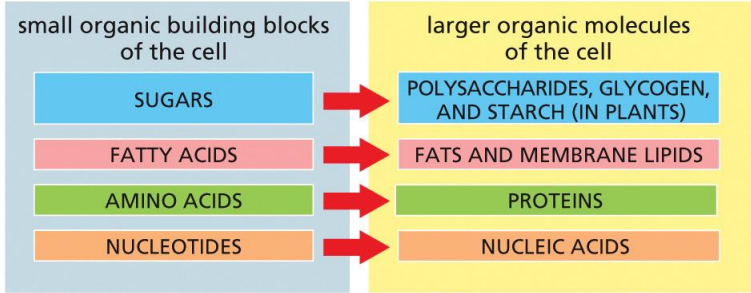
AN AMINO ACID



A FATTY ACID



A NUCLEOTIDE



SUBUNIT

MACROMOLECULE



sugar



polysaccharide



amino acid



protein

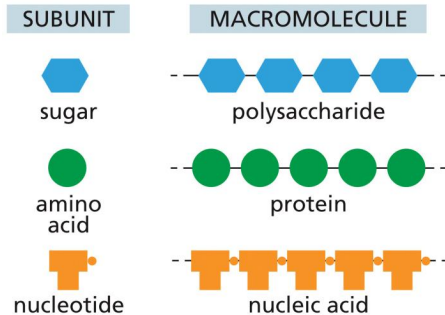
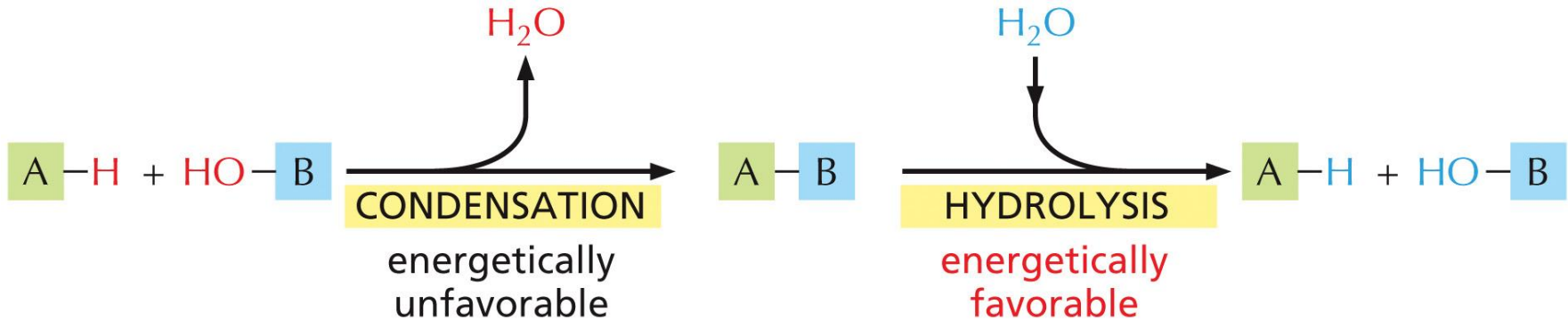


nucleotide



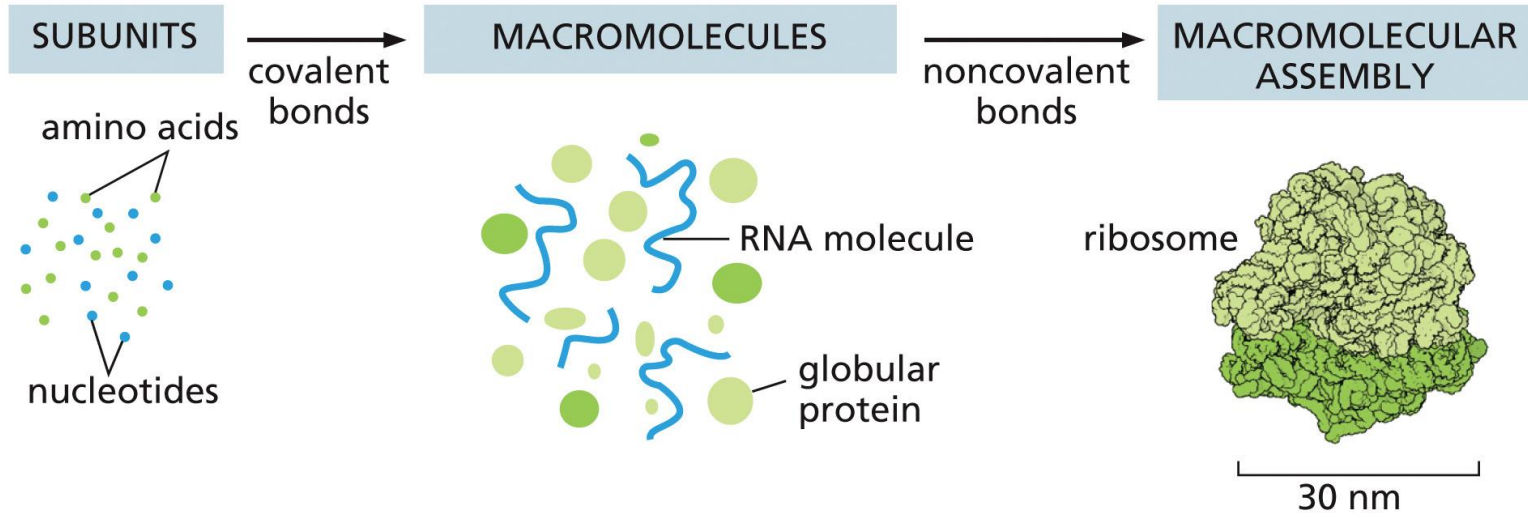
nucleic acid

Condensation and hydrolysis as opposite reactions



Condensation requires energy!

small molecules are **covalently linked** to form macromolecules
macromolecules assemble through **noncovalent** interactions forming large complexes



→ noncovalent forces drive protein folding and enzymatic activity

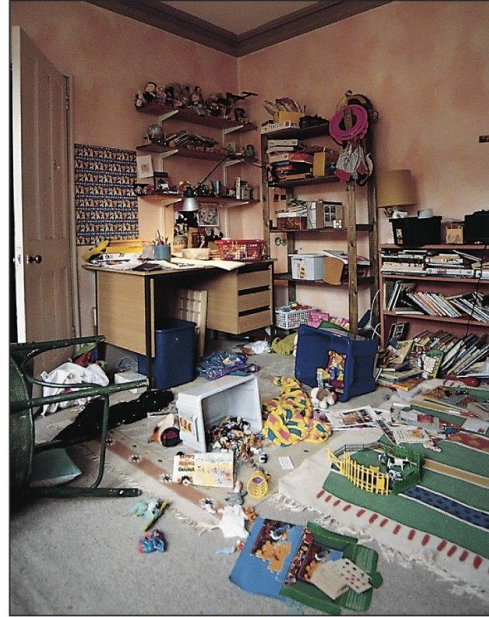
How do cells create order in a chaotic universe?



2^o law of thermodynamics: **disorder always increases**

"SPONTANEOUS" REACTION

as time elapses



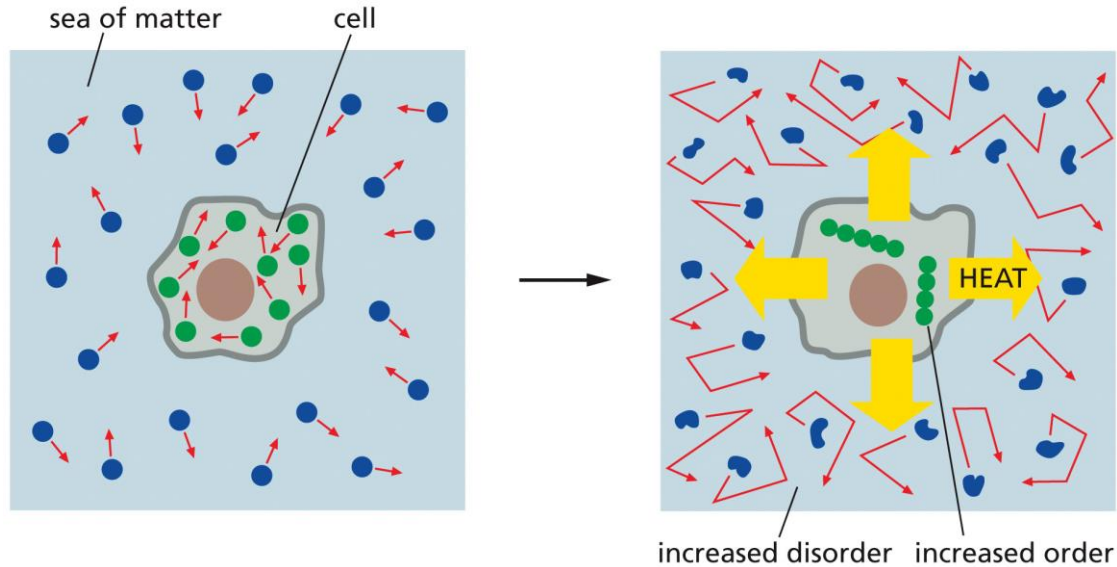
ORGANIZED EFFORT REQUIRING ENERGY INPUT

Do cells defy the law?

Erasmus MC

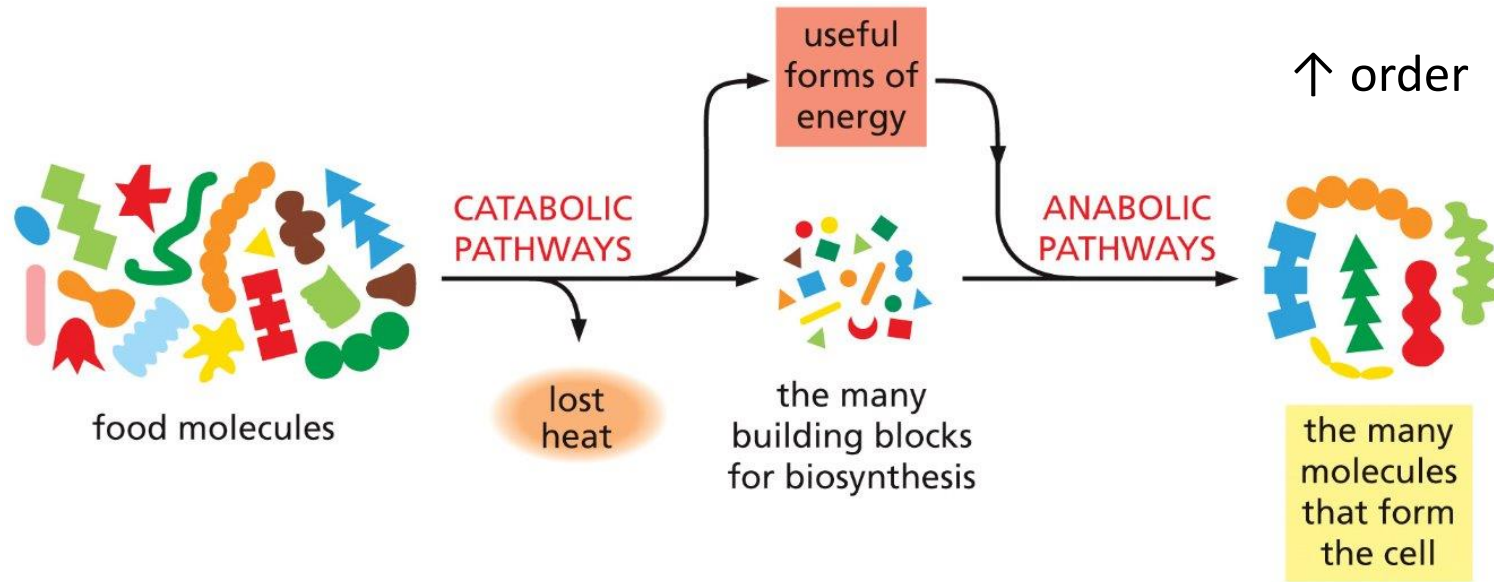
Erasmus

cells take energy from food to order molecules, but they release heat creating increased motion $\rightarrow$ more disorder overall



heat is energy in its most disordered form

Cell metabolism: coordination of catabolic and anabolic reactions

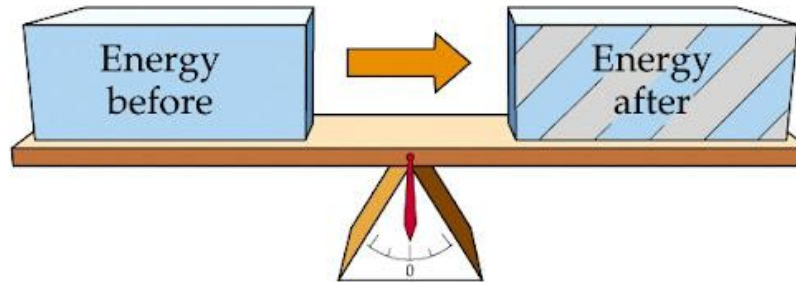


→ altogether = metabolism, made possible by enzymes

1^o law of thermodynamics: **energy is not created or destroyed**

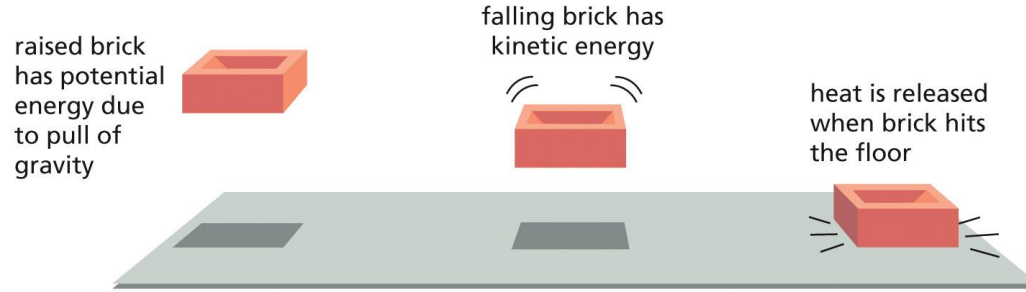
(a) The First Law of Thermodynamics

Energy
transformation

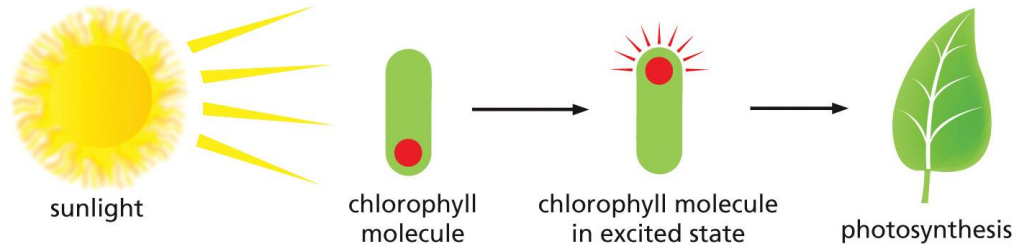


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→ total energy is always conserved

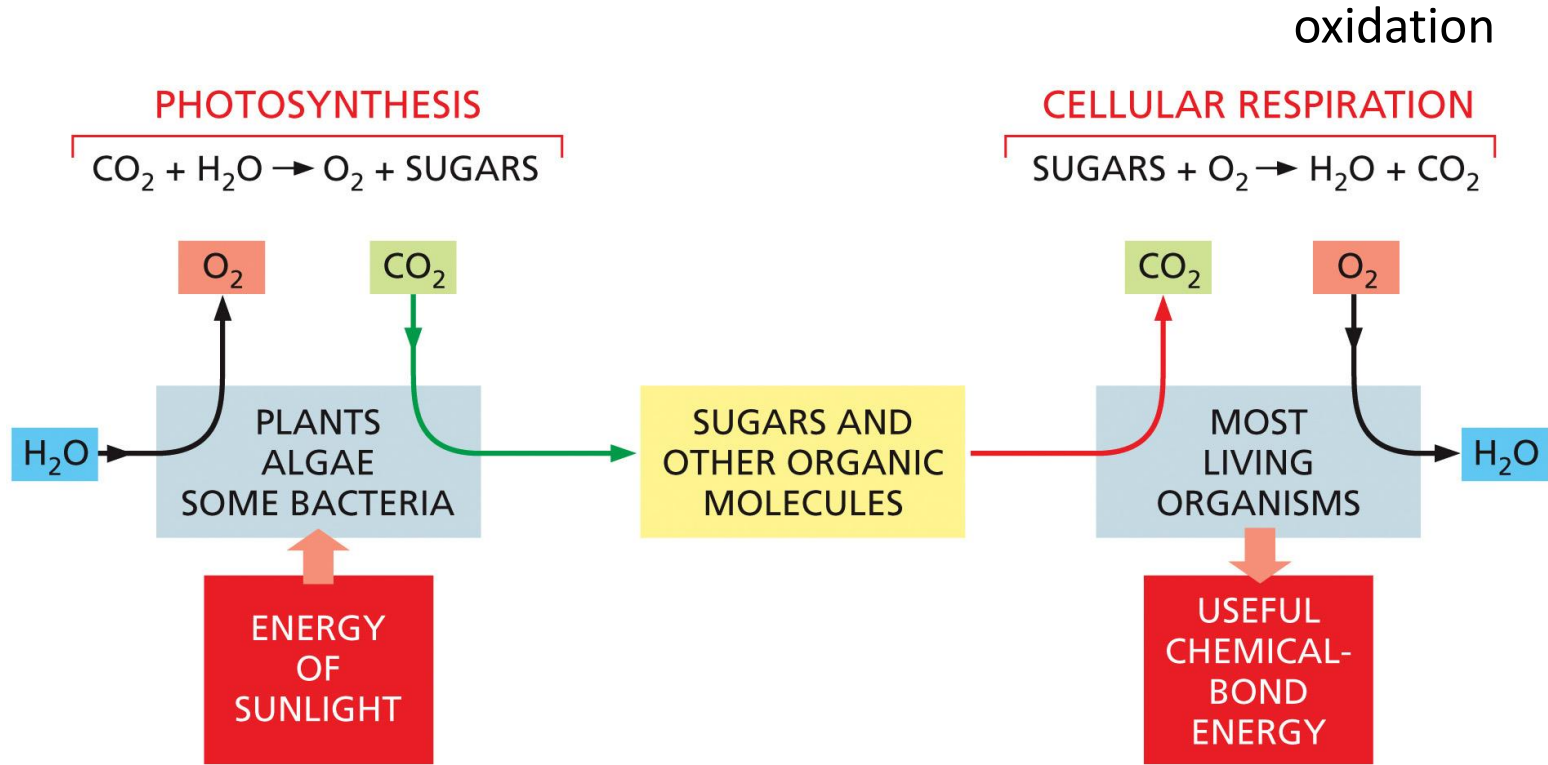


A potential energy due to position → kinetic energy → heat energy



D electromagnetic (light) energy → high-energy electrons → chemical-bond energy

Carbon atoms cycle through the biosphere

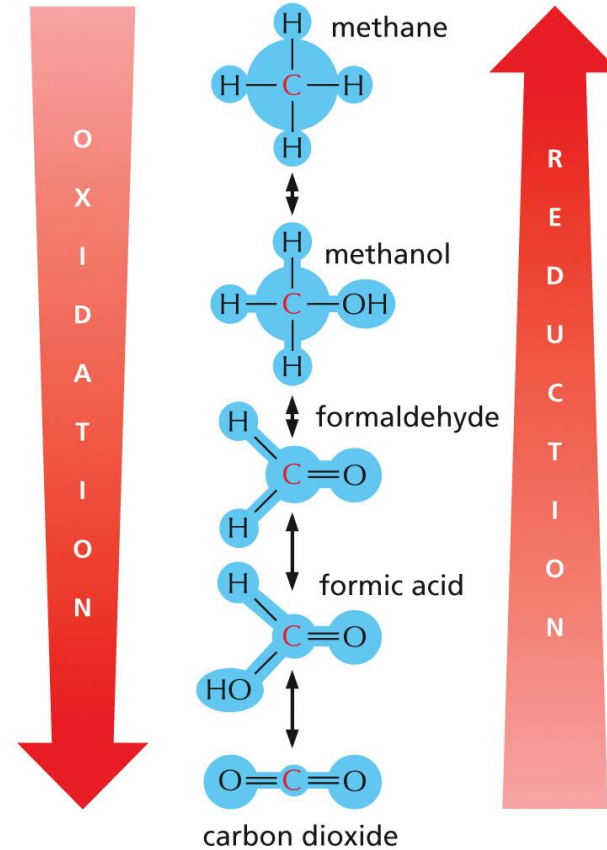


oxidation: removal of electrons

→ H atoms replaced by O atoms

→ decreased number C-H bonds

*reduction & oxidation (“redox”)
always occur simultaneously
→ conservation nr. electrons*



Favorable reactions lead to a reduction in free energy (G)

G = energy available “to do work”
(e.g. drive chemical reactions)

reduction of G ($\Delta G < 0$): **reduction of order**

increase of G ($\Delta G > 0$): **increase of order**

ENERGETICALLY
FAVORABLE
REACTION



The free energy of Y is greater than the free energy of X. Therefore $\Delta G < 0$, and the disorder of the universe increases during the reaction $Y \rightarrow X$.

ENERGETICALLY
UNFAVORABLE
REACTION



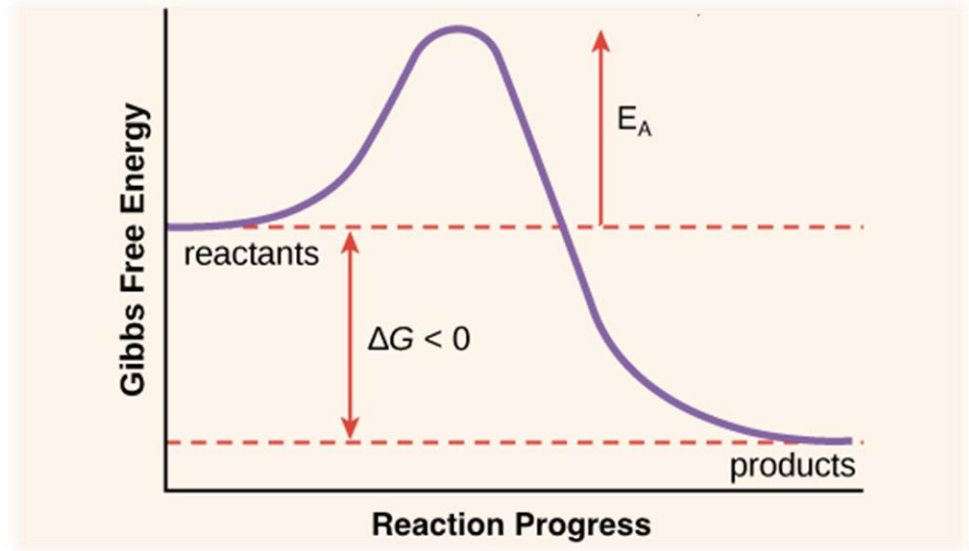
If the reaction $X \rightarrow Y$ occurred, ΔG would be > 0 , and the universe would become more ordered.

Activation energy = barrier for the reaction to proceed

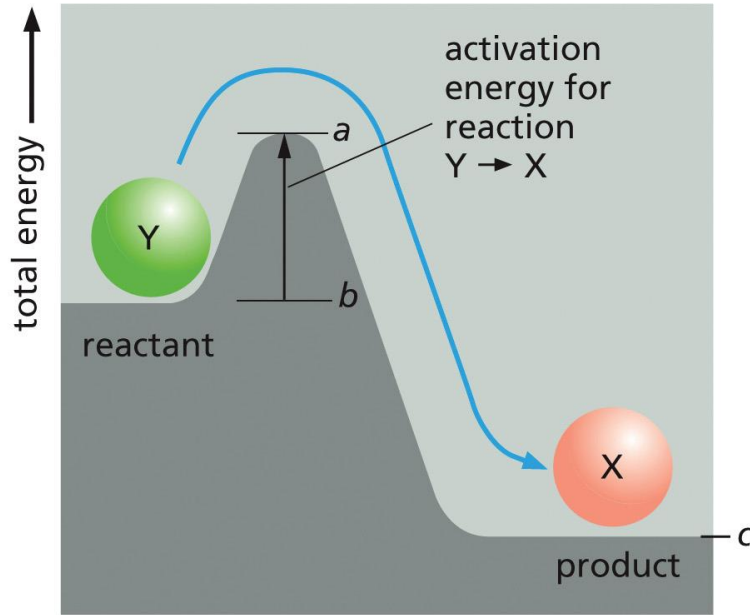
Favorable reaction (G decreases)



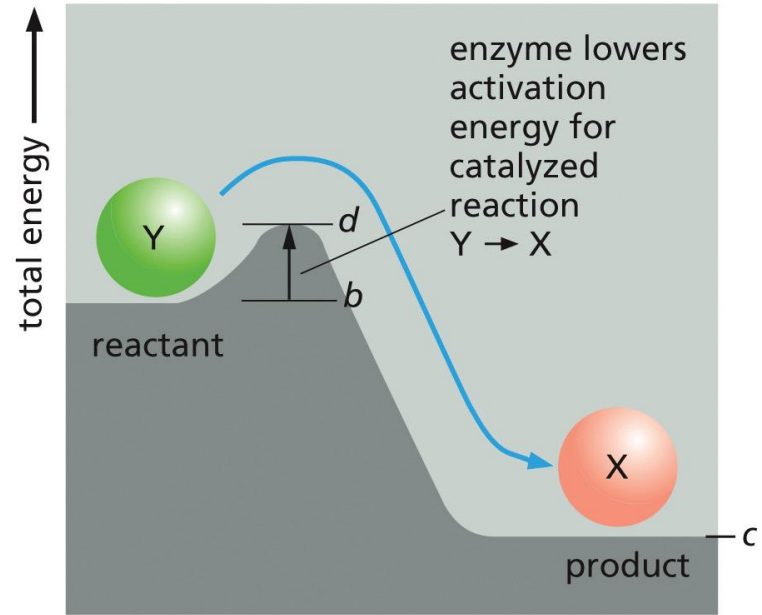
*heat of a lighted match,
collision between molecules...*



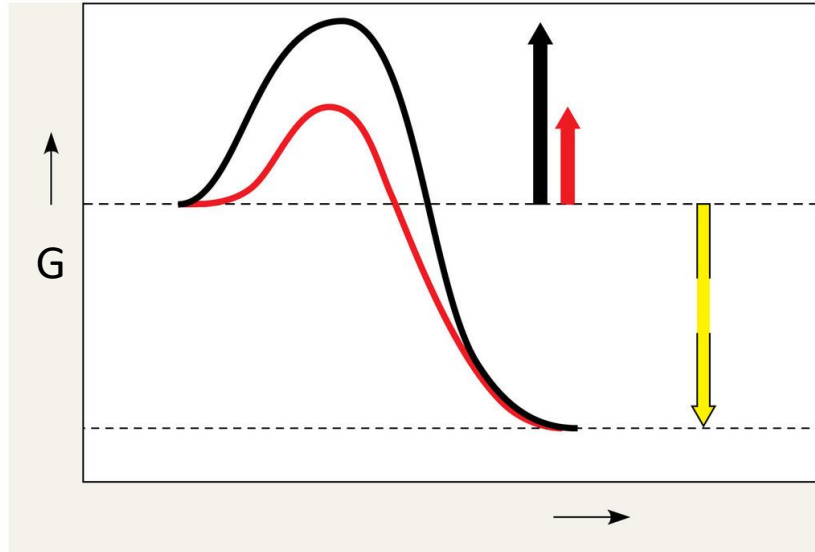
Enzymes lower the activation energy



(A) uncatalyzed reaction pathway



(B) enzyme-catalyzed reaction pathway



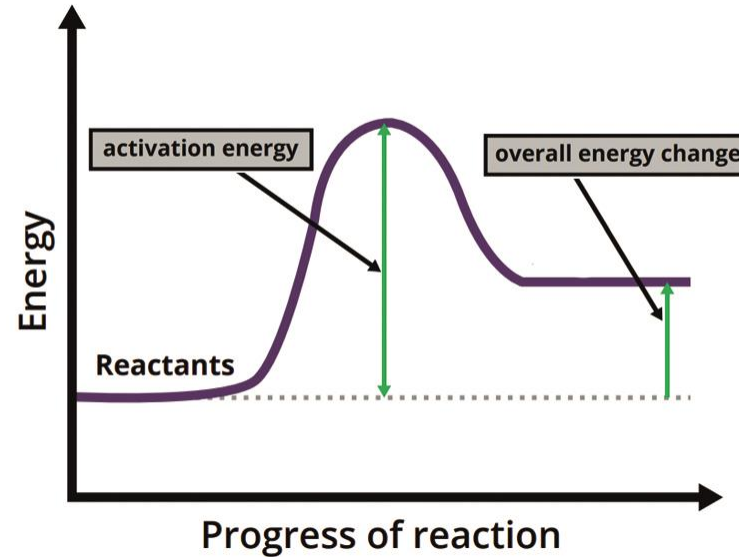
1) The graph shows a reaction with a reduction in free energy G . This could represent:

- A. Addition of an amino acid during the synthesis of β -globin
- B. Photosynthesis in *aloe vera*
- C. Hydrolysis of the complex sugar glycogen to glucose (monomer)
- D. A reaction that releases heat
- E. All of the previous

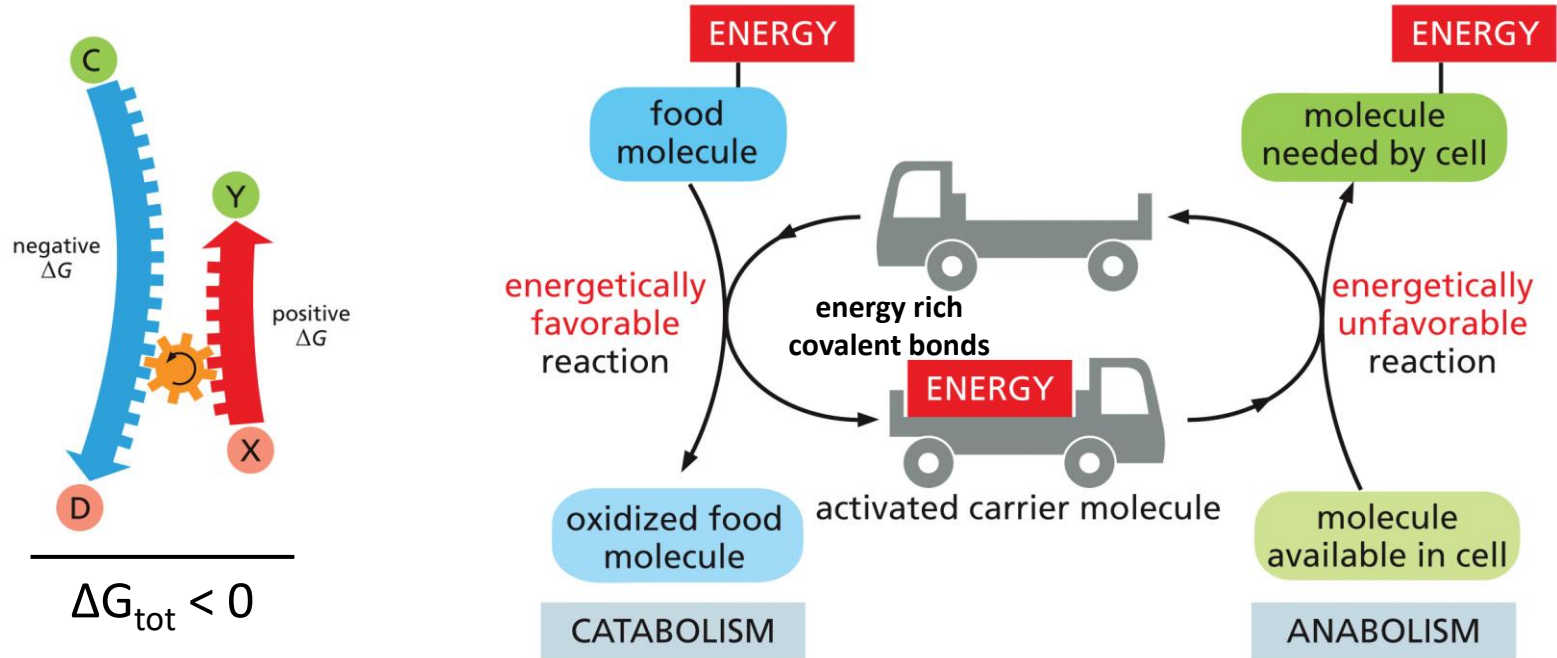
2) This reaction is “spontaneous” since:

- A. It requires no energy input to occur
- B. It happens very quickly
- C. It cannot be stopped or slowed down
- D. It can also occur without enzymes
- E. None of the previous

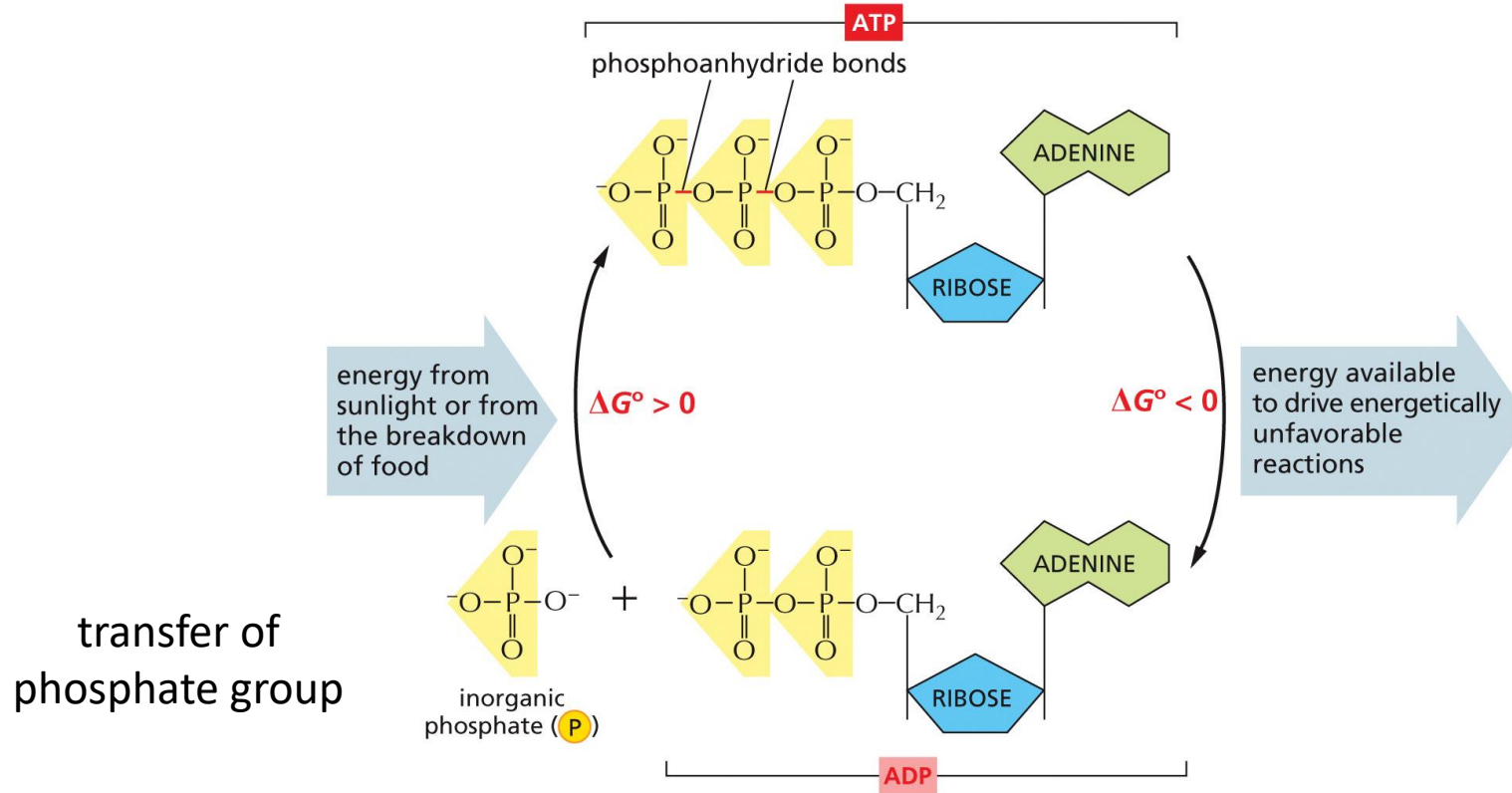
...but many important reactions
in the cell are unfavourable!



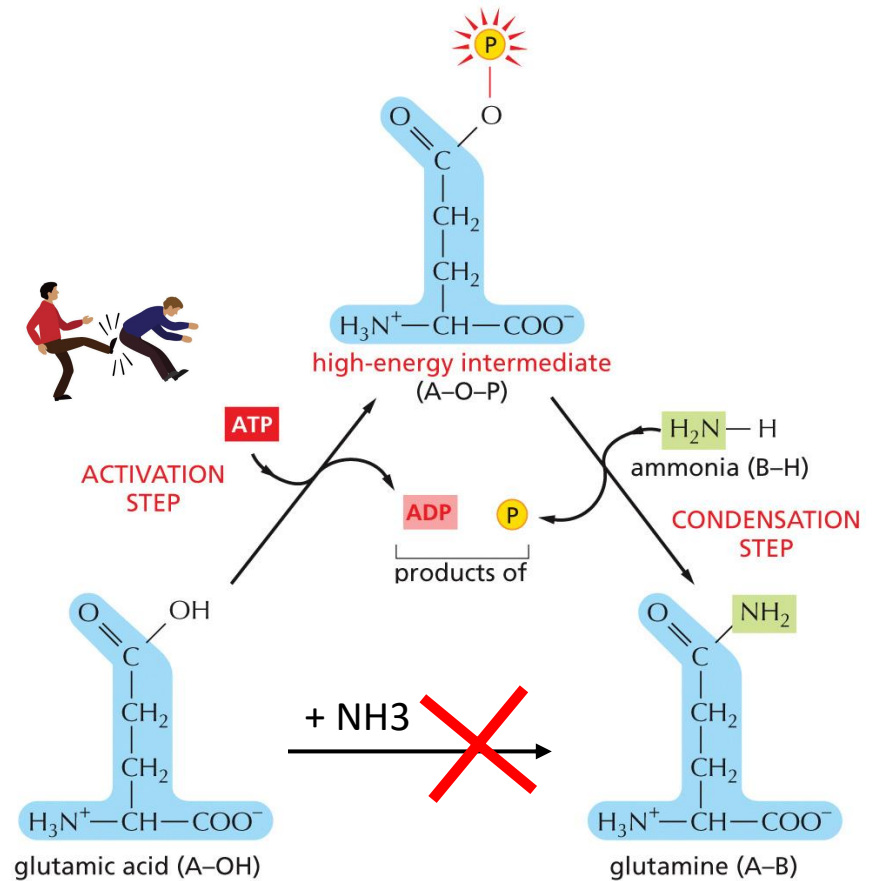
Driving unfavorable reactions by reaction coupling



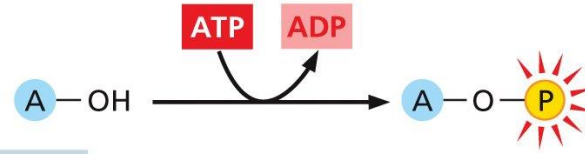
1. **ATP**: the most used activated carrier molecule



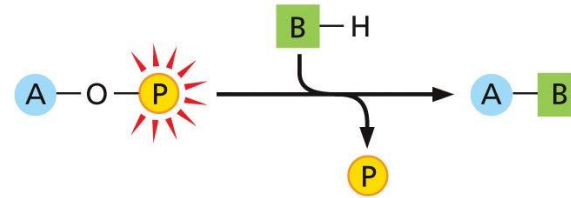
ATP-derived energy is often used to join molecules together



ATP-derived energy is often used to join molecules together (including macromolecules)



STEP 1 in the **ACTIVATION** step, ATP transfers a phosphate, **P**, to A-OH to produce a high-energy intermediate



STEP 2 in the **CONDENSATION** step, the activated intermediate reacts with B-H to form the product A-B, a reaction accompanied by the release of inorganic phosphate

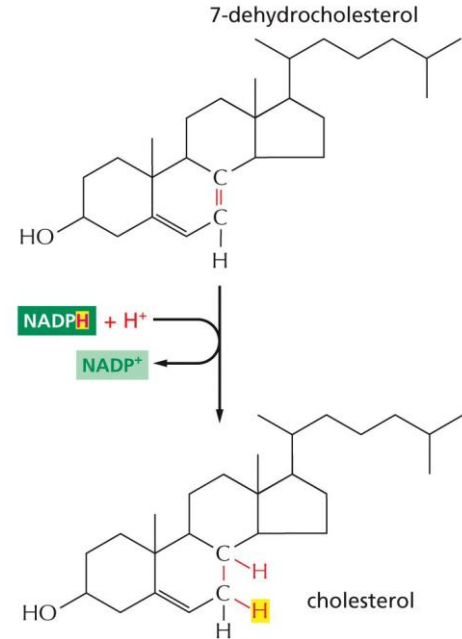
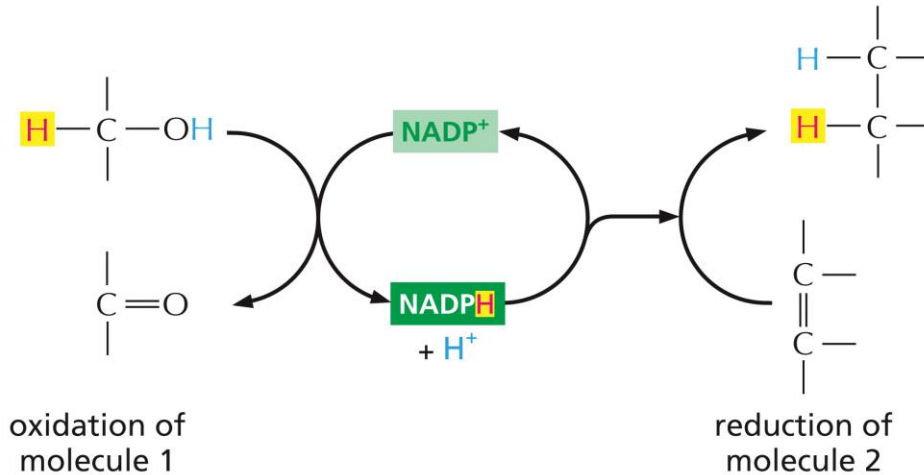
NET RESULT



2 NADH & NADPH: electron carriers in ox-red reactions

oxidised NAD(P)H releases e^- to power unfavorable reactions

uptake of two e^- and one H^+



3. **Acetyl-CoA** provides two C in the biosynthesis of large molecules

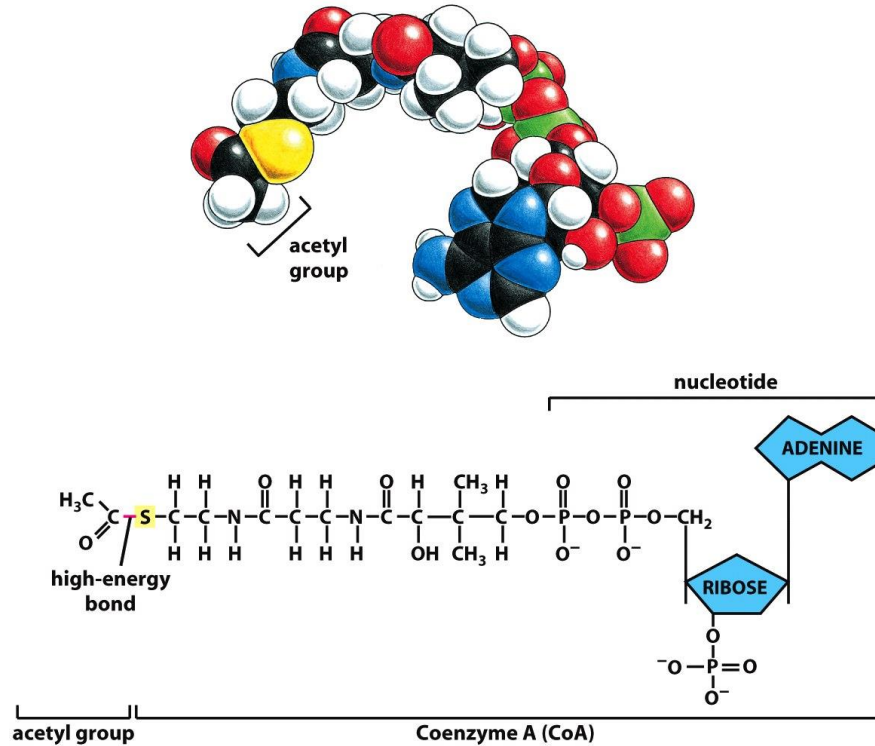
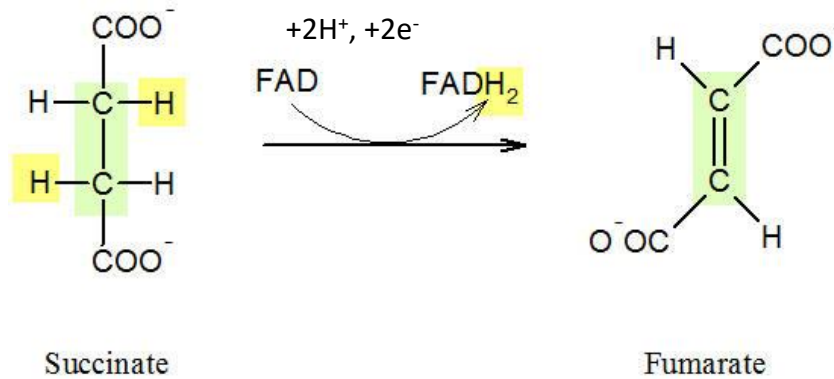


TABLE 2–3 Some Activated Carrier Molecules Widely Used in Metabolism

Activated carrier	Group carried in high-energy linkage
ATP	Phosphate
NADH, NADPH, FADH ₂	Electrons and hydrogens
Acetyl CoA	Acetyl group
Carboxylated biotin	Carboxyl group
S-Adenosylmethionine	Methyl group
Uridine diphosphate glucose	Glucose



1) Succinate dehydrogenase is an enzyme which catalyses the conversion of succinate to fumarate during the digestion of fat. The energy carrier FAD is converted to FADH₂ in the process. Which of the following statements are true?

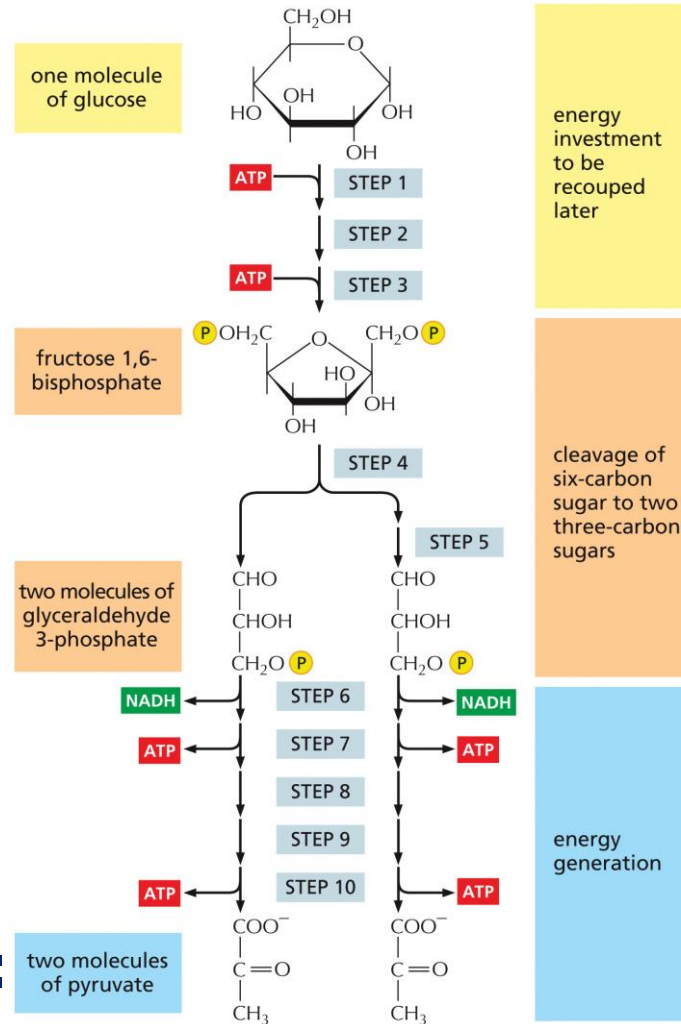
- A. It is an oxidation reaction since both succinate and FAD are oxidised
- B. Succinate is oxidised while FAD is reduced
- C. The reaction has negative ΔG
- D. The reaction has positive ΔG
- E. FAD can be later regenerated by reducing FADH₂

Cells obtain energy from food

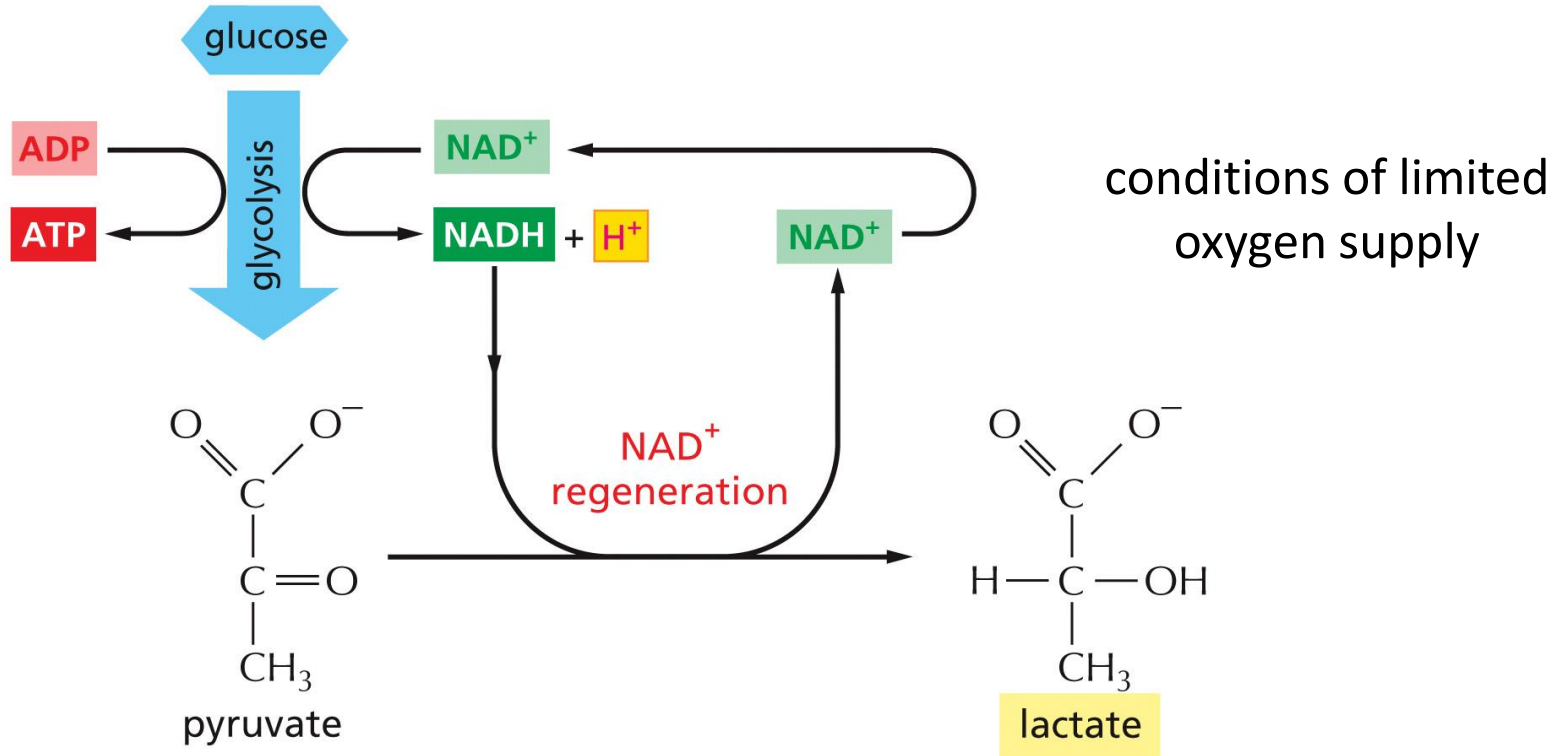
Anaerobic glycolysis:
glucose $\rightarrow$ 2x pyruvate

2 ATP consumed
4 ATP, 2 NADH generated

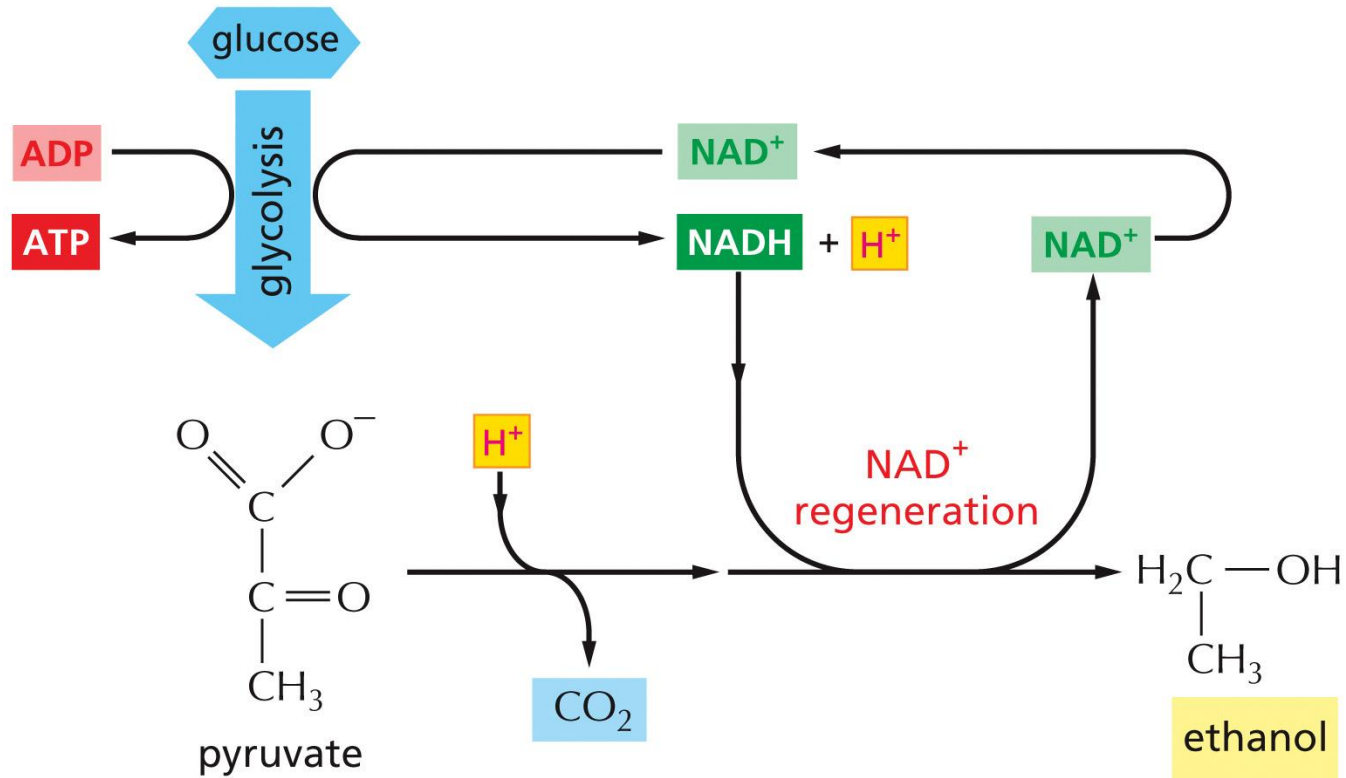
The fate of pyruvate
depends on O_2 availability



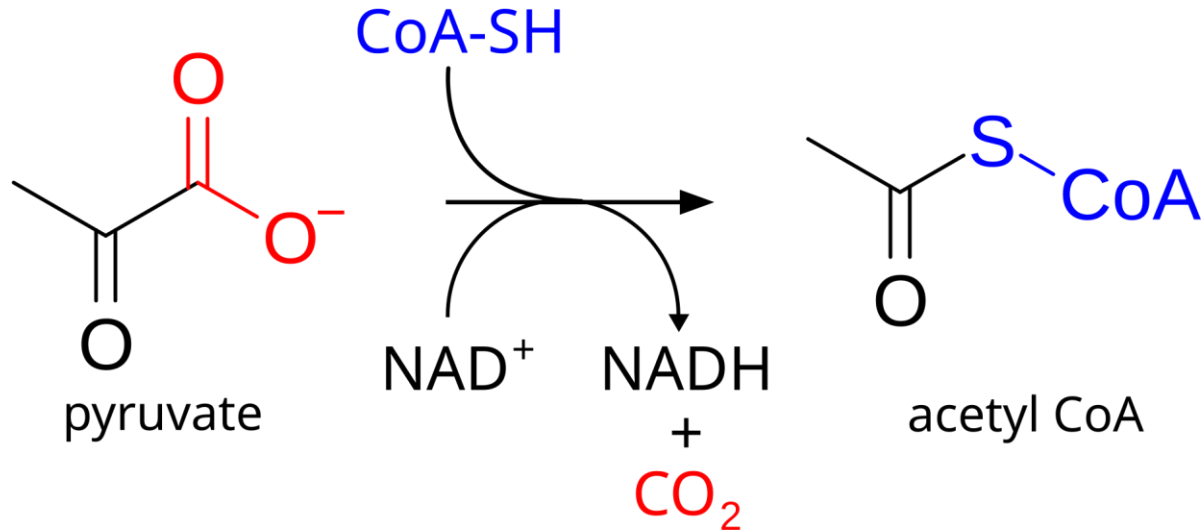
FERMENTATION IN A VIGOROUSLY ACTIVE MUSCLE CELL



FERMENTATION IN YEAST

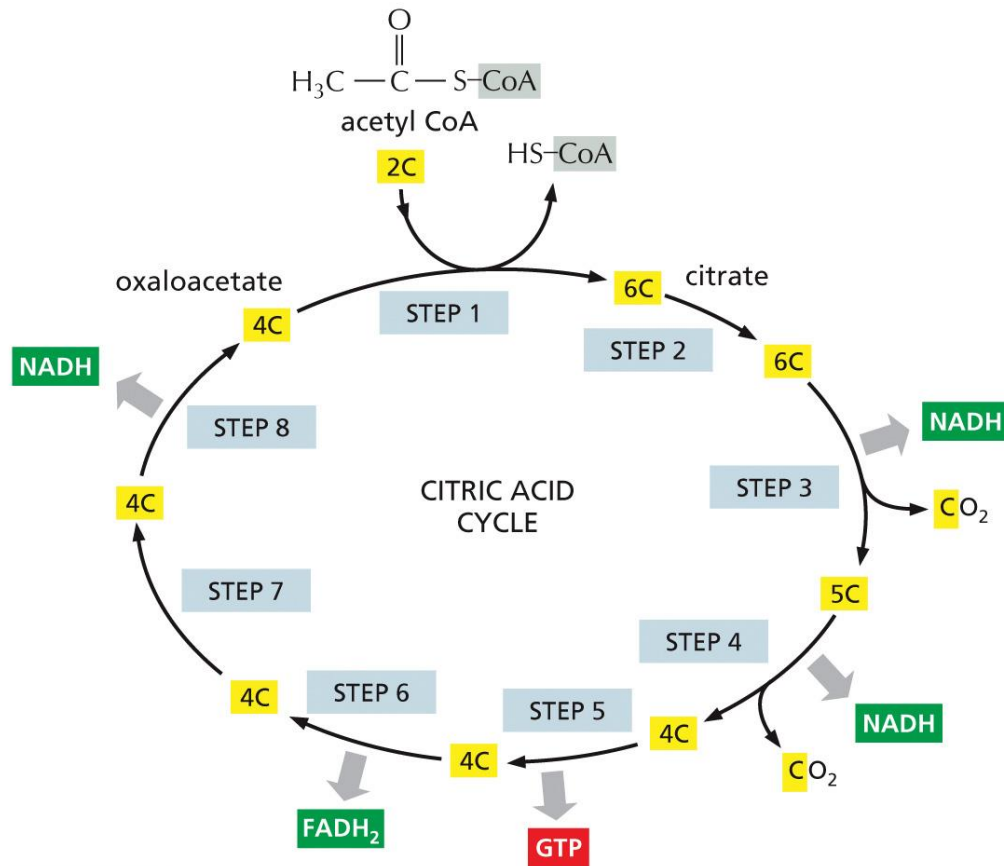


if oxygen is present, pyruvate enters mitochondria



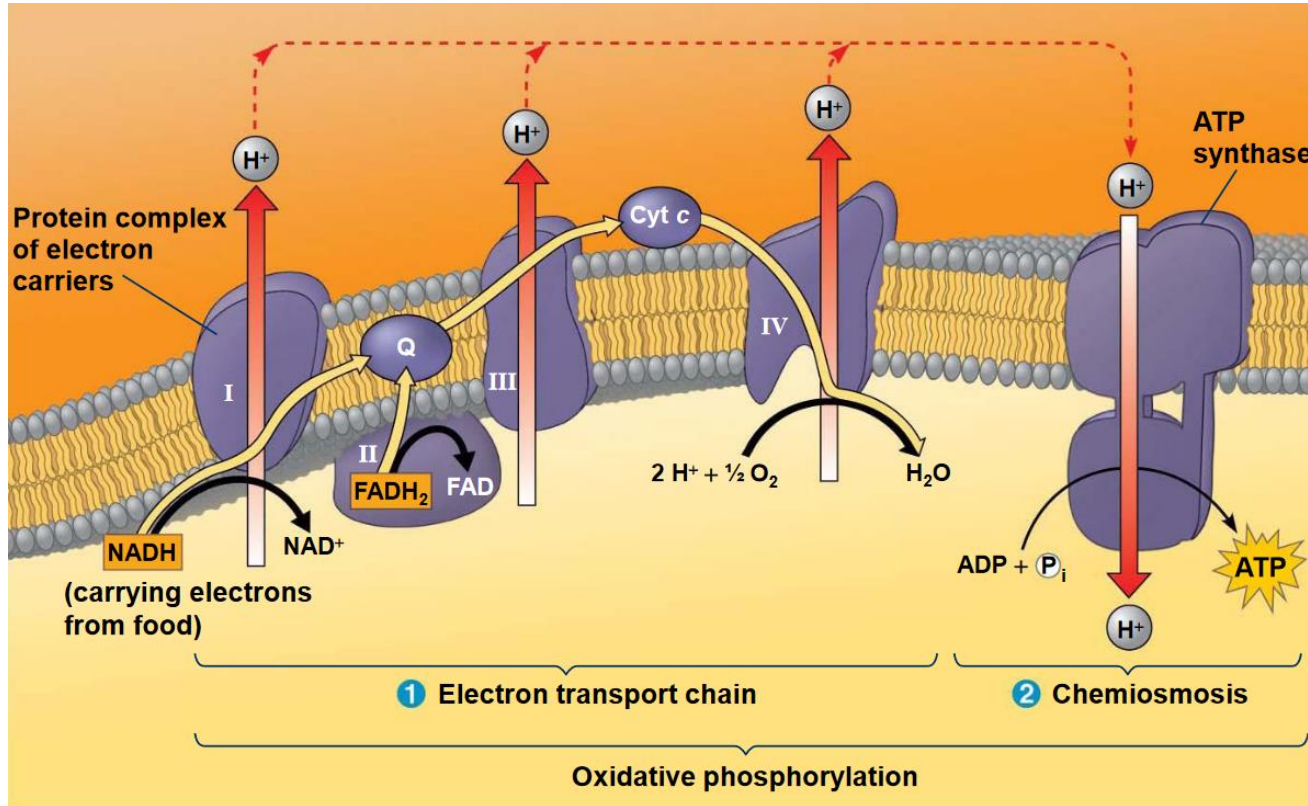
Citric acid cycle in mitochondria

aminoacids and fatty acids can also enter the cycle as acetyl CoA or other intermediates

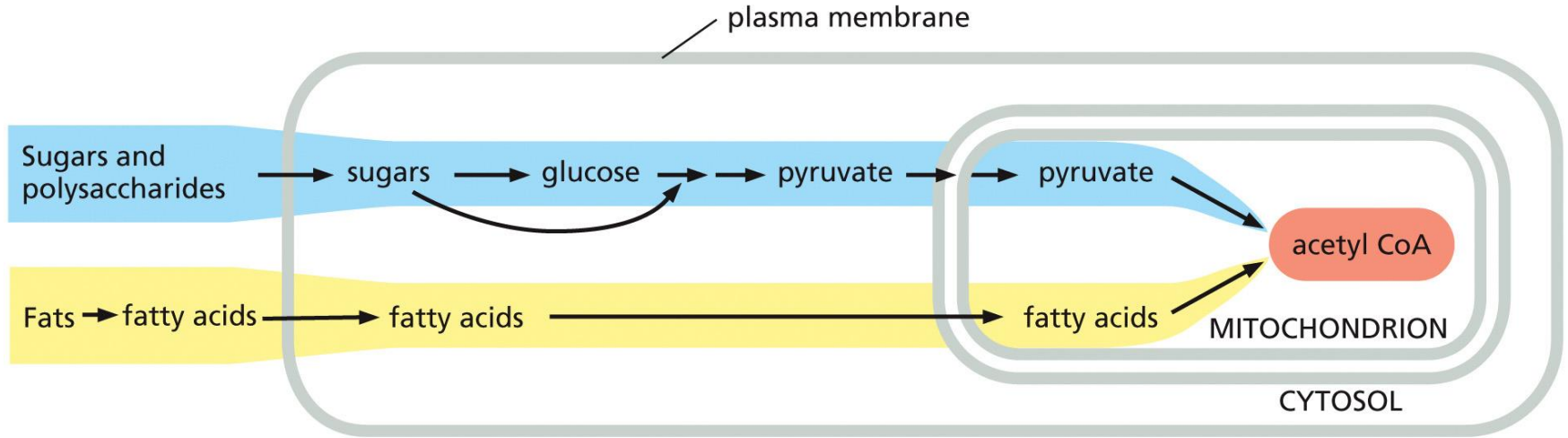


NET RESULT: ONE TURN OF THE CYCLE PRODUCES THREE NADH, ONE GTP, AND ONE FADH₂, AND RELEASES TWO MOLECULES OF CO₂

In the next episodes... conversion of electron energy to ATP



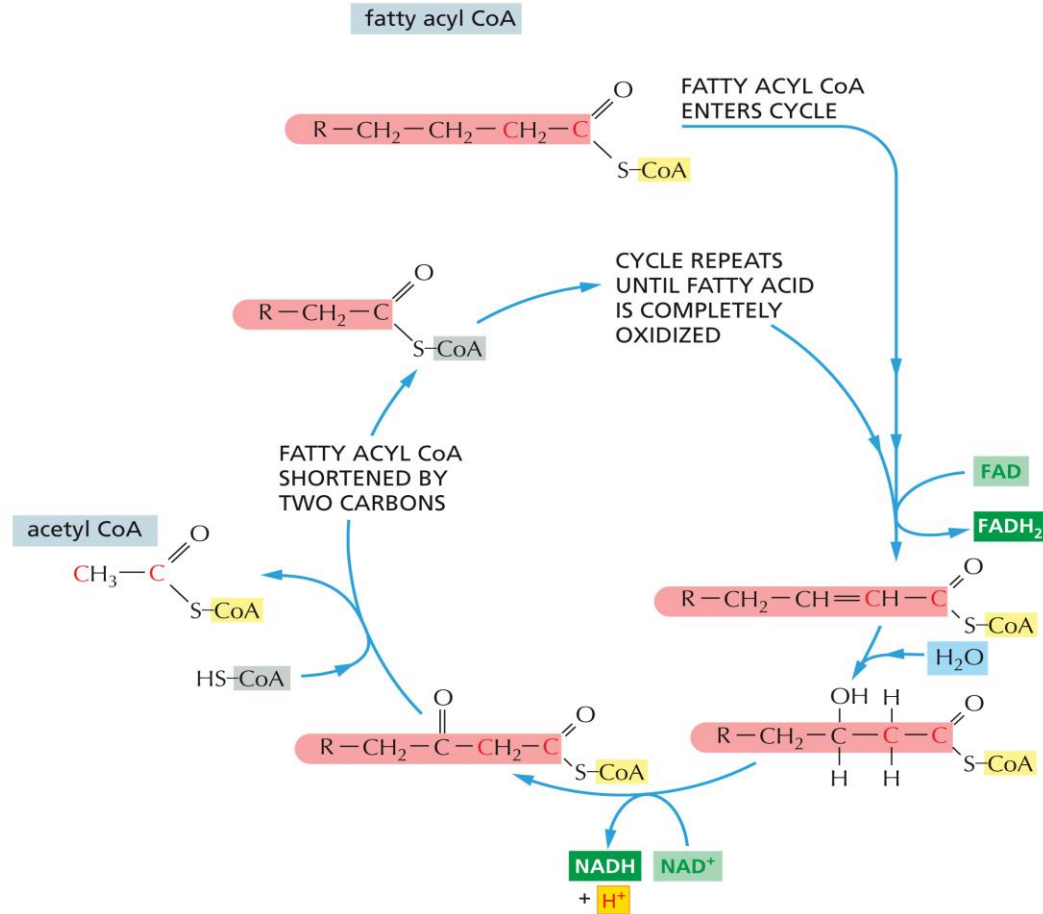
Pyruvate and FAs are moved into mitochondria and oxidised

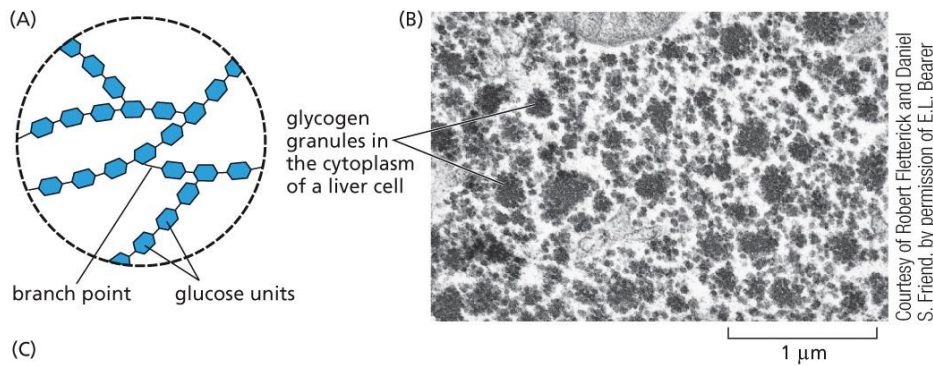


Fatty acyds are oxidized
to acetyl CoA through a
metabolic cycle in
mitochondria

2 C removed/cycle

acetyl CoA can enter the
citric acid cycle

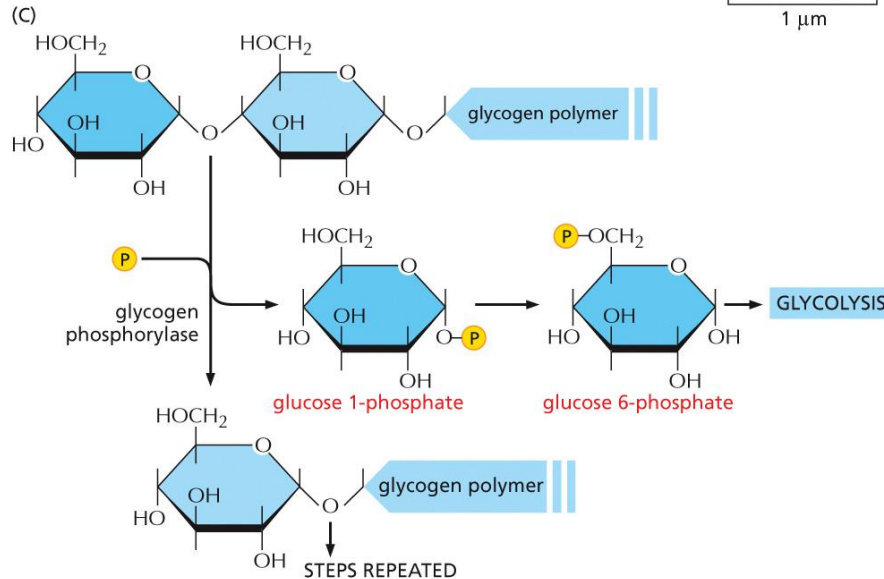




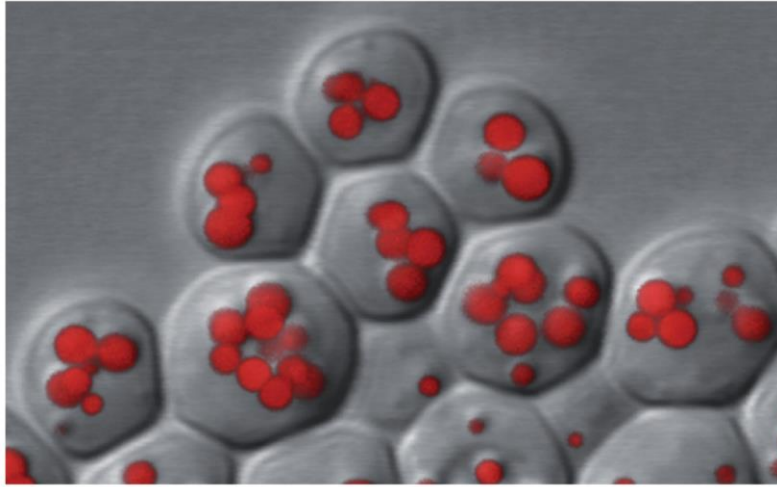
Storage of energy: glycogen

sufficient for approx 1
day of normal activity

sugars (glucose) are
produced from glycogen

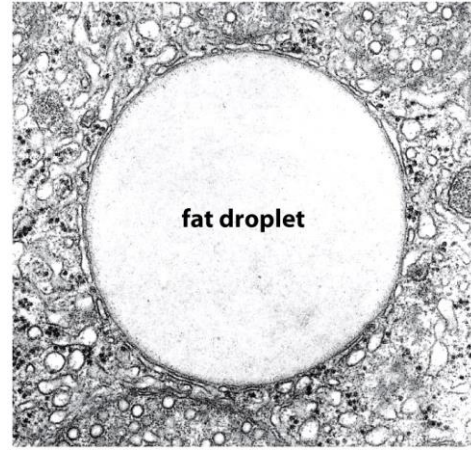


Storage of energy: **fat**

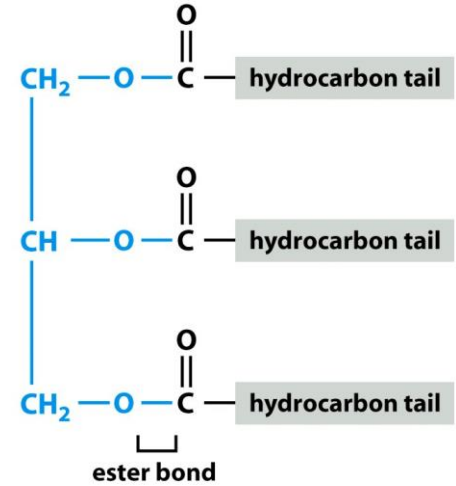


Courtesy of Sepp D. Kohlwein

10 μm

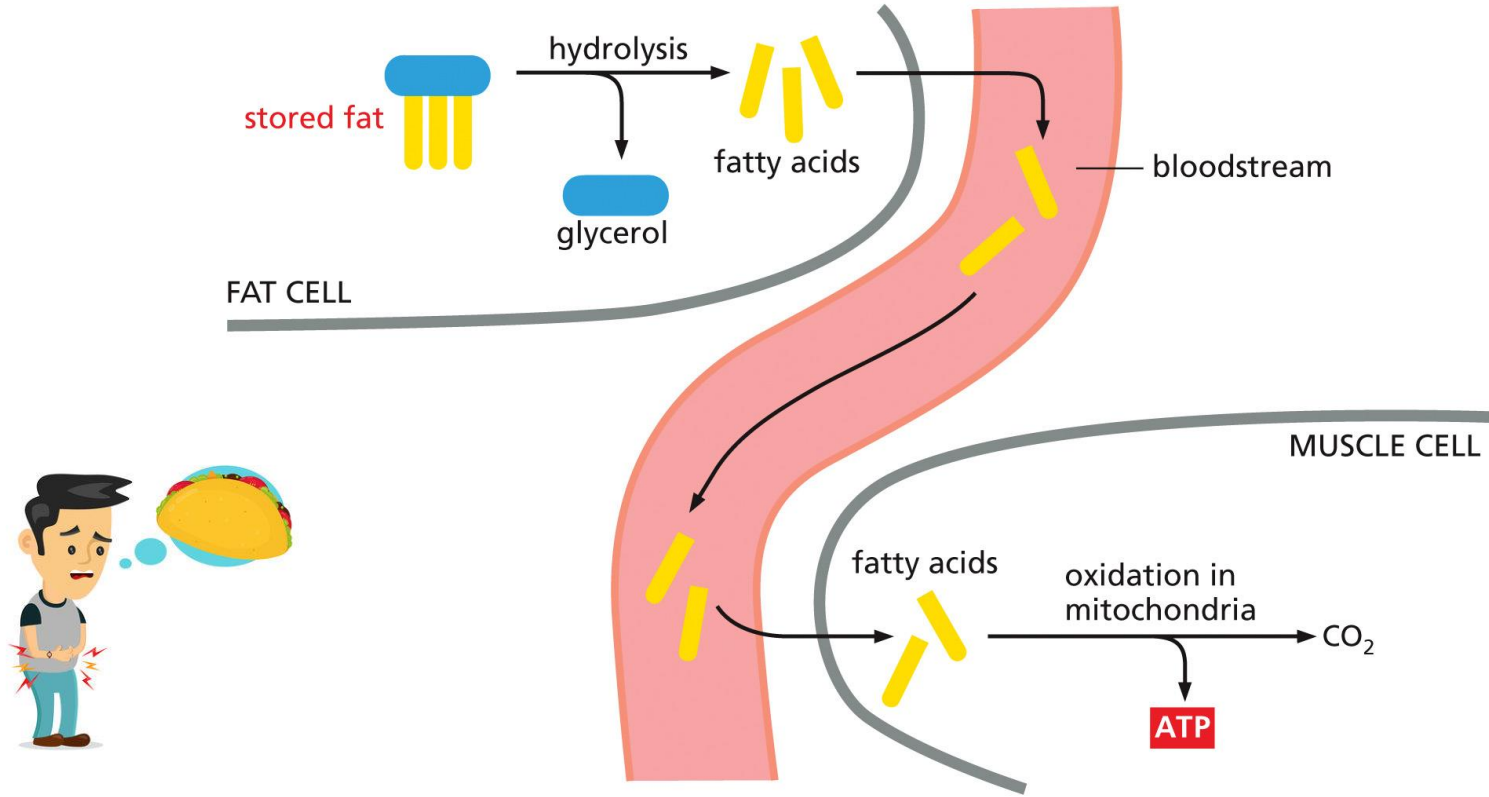


1 μm

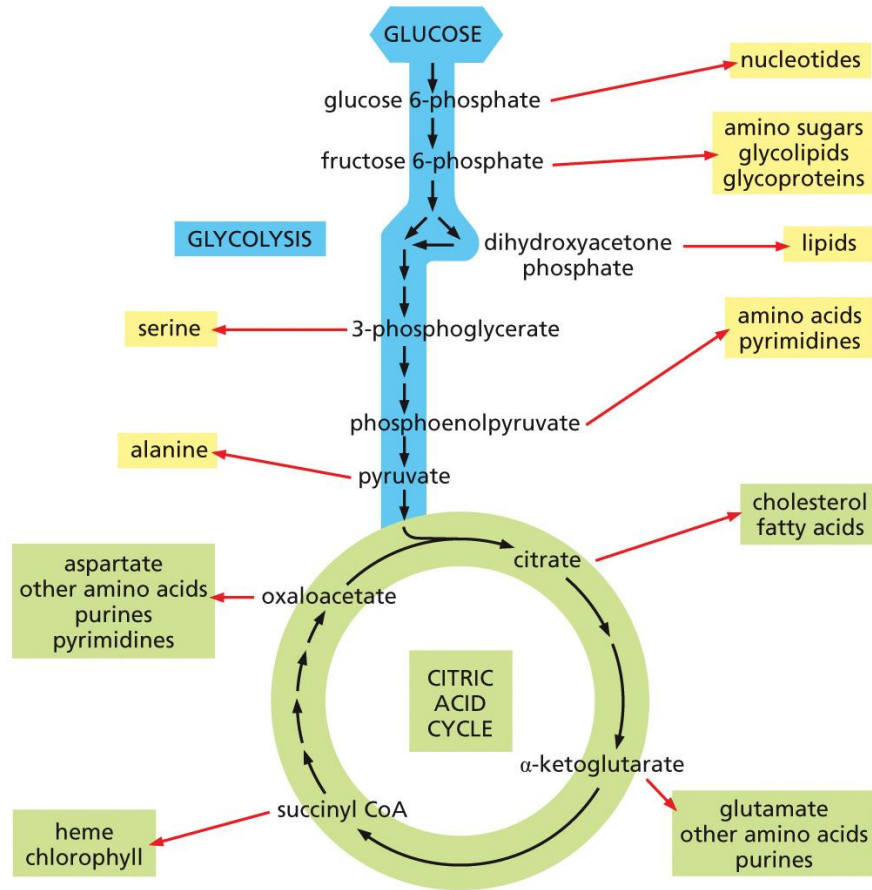


sufficient for approx 1
month of normal activity

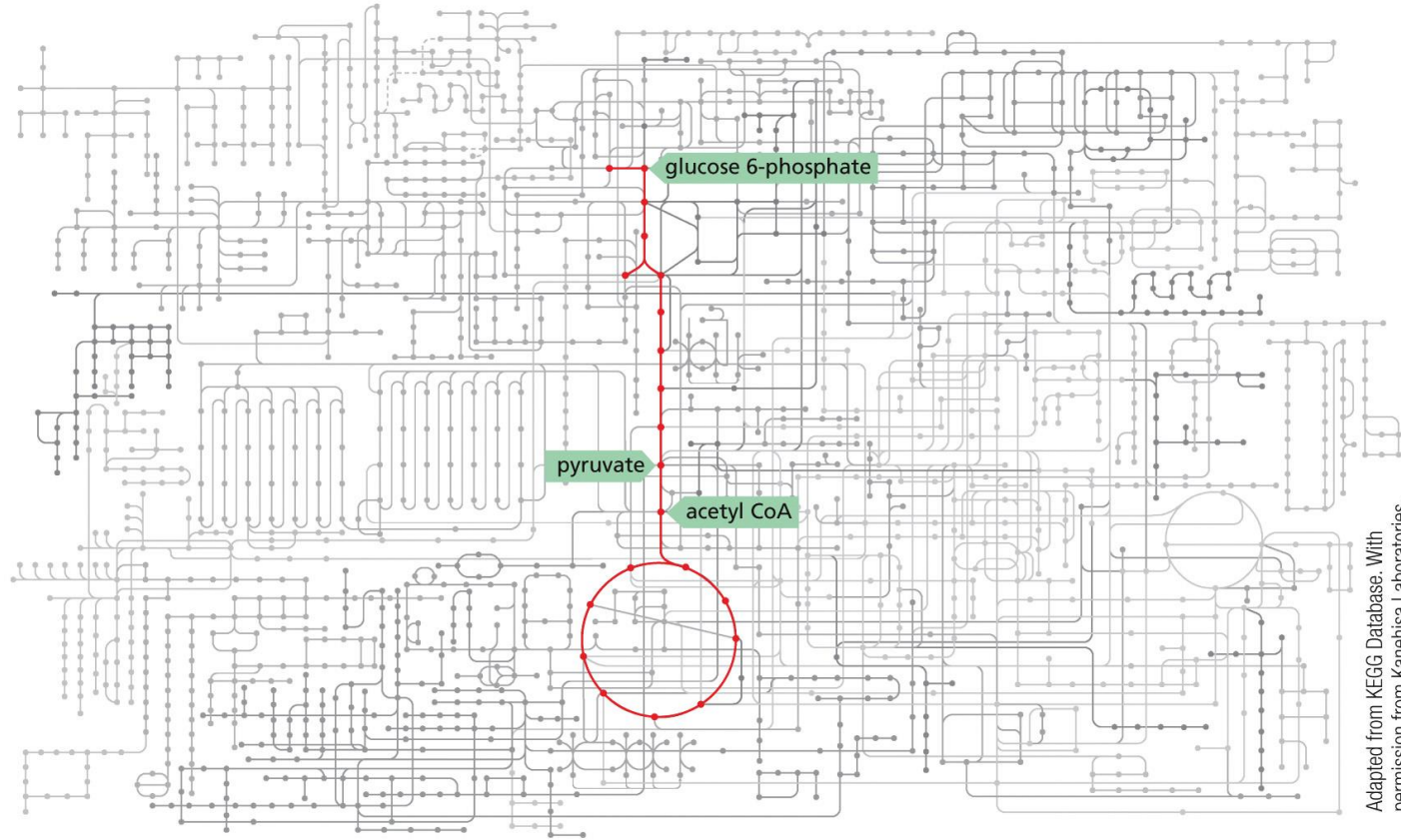
Mobilisation of stored fat for energy production



Glycolysis and citric acid cycle provide molecules for biosynthesis in addition to energy



A more realistic (still simplified!) view of metabolic pathways



Adapted from KEGG Database. With permission from Kanehisa Laboratories.

Key points

- ✓ biomolecules → polymers → cost energy ($\Delta G > 0$)
- ✓ enzymes lower the activation energy and energy-carriers store energy derived from food
- ✓ food: sugars → **glycolysis** (cytoplasm)
fat → **fatty acids cycle** (mitochondria)
- ✓ in mitochondria: acetyl CoA → **citric acid cycle** and oxidative phosphorylation (ATP production)
- ✓ intermediates from glycolysis/citric acid cycle are also precursors for biosynthesis